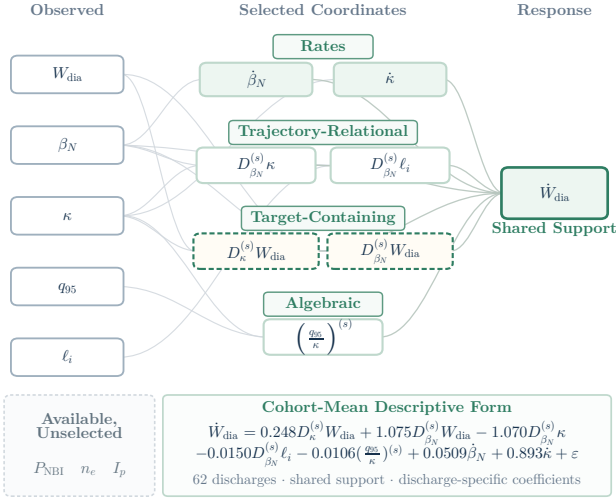


DIII-D: Task-Conditioned Relational Discovery

One 62-discharge observational record $\rightarrow$ Distinct qualified organizations under q_{desc} and q_{rec}

a Descriptive Relational Organization (q_{desc})

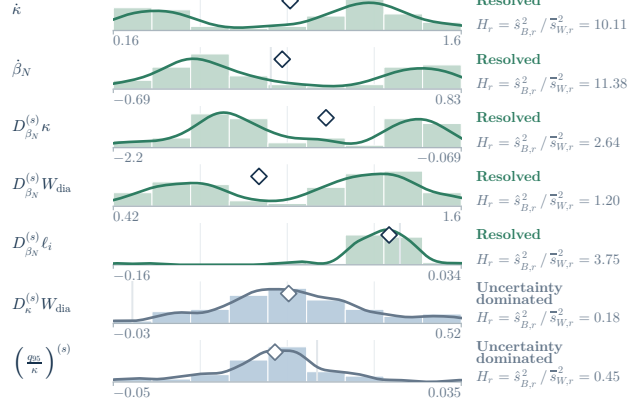


b Coefficient Heterogeneity Audit (q_{desc})

62 discharge-specific coefficients

Decision rule

Resolved when between-discharge variation persists beyond within-discharge uncertainty



Native-scale histogram + KDE; range: observed min-max; diamond: random-effects mean μ_{REML} .

$D_{\beta_N}\ell_i$ is resolved ($H_r = 3.75$, $\tau_{REML}^2 > 0$);

$D_{\kappa}W_{dia}$ is uncertainty-dominated ($H_r = 0.18$, $\tau_{REML}^2 = 0$).

c Recurrent Support Terms Across Folds (q_{rec})

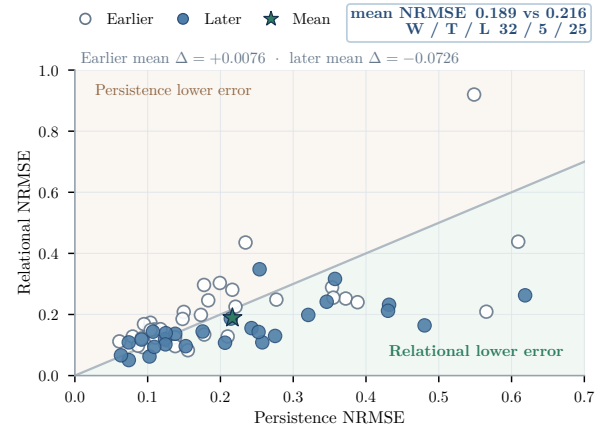
Type	Relational Term	F1	F2	F3	F4	F5	F6	Rec.
ratio	ece21 / cerTi10	█	█	█	█	█	█	6
id	W_{dia}	█	█	█	█	█	█	6
ratio	$t_{inj} / ece20$	█	█	█	█	█	█	5
ratio	$n_{e,ped} / cerTi3$	█	█	█	█	█	█	5
recip	$1 / cerTi10$	█	█	█	█	█	█	4
ratio	Do03 / cerTi3	█	█	█	█	█	█	3
ratio	$I_p / ece39$	█	█	█	█	█	█	3
prod	$n_{e,ped} \times cerTi6$	█	█	█	█	█	█	3
ratio	I_p / B_t	█	█	█	█	█	█	3
recip	$1 / B_t$	█	█	█	█	█	█	3
prod	ece37 × ece39	█	█	█	█	█	█	3
ratio	$I_p / cerTi10$	█	█	█	█	█	█	2
prod	Do04 × cerV8	█	█	█	█	█	█	2
ratio	$P_{33R} / cerTi10$	█	█	█	█	█	█	2
prod	$P_{inj} \times ece36$	█	█	█	█	█	█	2

Legend: Ratio (green), Product (purple), Reciprocal (brown), Identity (dark blue)

Rows show 15 terms recurring in at least two of six 12-term supports
20 singleton terms are omitted.

Ratio constructions comprise 8 / 15 recurrent terms. Ratio and reciprocal terms shown passed the frozen admissibility and range-support checks in every fold where selected.

d Target-Cross-Fitted Density Reconstruction (q_{rec})



62 out-of-fold discharges · finite-object, target-cross-fitted qualification
not external validation